

Chapter 5A: Algebra

Drill practice and mixed revision of all 8 lessons · 77
questions

NSW syllabus outcome

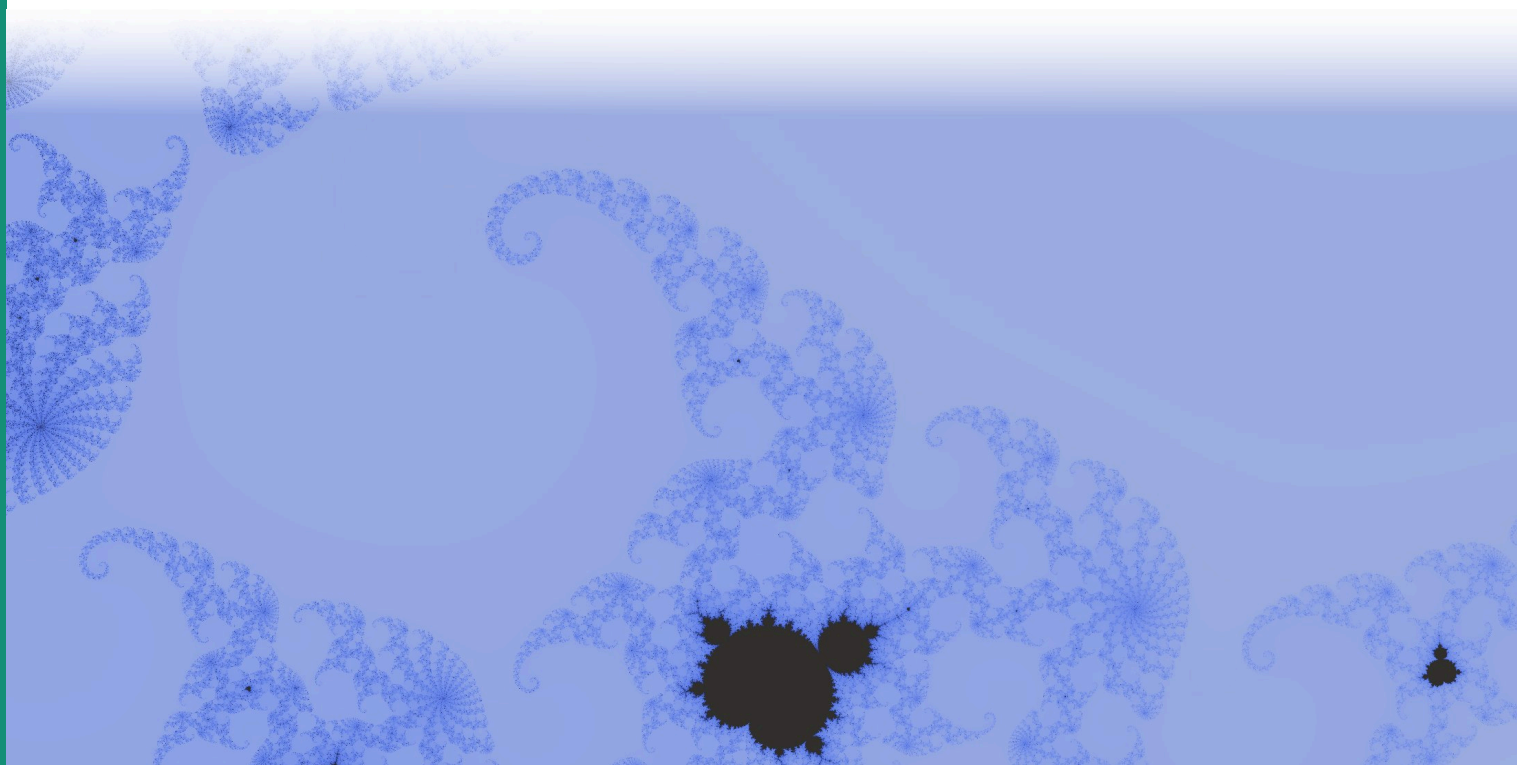
Stage 4 Number and algebra: MA4-ALG-C-01 generalises
number properties to operate with algebraic expressions.

NAME

CLASS

CHAPTER

5A



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 5A. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1-35

5.01 The commutative and associative laws

Easy Evaluate mentally. Pair the friendly numbers first.

1 $18 \times 125 \times 8$

2 $34 \times 50 \times 2$

3 $25 \times 28 \times 4$

4 $39 \times 2 \times 50$

Medium Evaluate mentally. Rearrange to make tens first.

5 $48 + 58 + 22$

6 $74 + 26 + 6$

7 $27 + 89 + 13$

8 $20 + 73 + 30$

5.02 The distributive law

Easy Evaluate by splitting the number: $6 \times 23 = 6 \times 20 + 6 \times 3$.

9 9×21

10 9×25

11 5×15

12 5×44

Medium Fill in the missing number.

13 $7 \times 61 = 7 \times \square + 7 \times 1$

14 $7 \times 38 = 7 \times 40 - 7 \times \square$

5.03 Variables and algebraic notation

Easy Simplify. Write the number first, then the letters in alphabetical order.

15 $a \times 12$

16 $m \times 3$

17 $b \times 6$

18 $6 \times b$

Medium Simplify.

19 $p \times y \times y$

20 $8 \times p \times p \times y$

21 $5 \times p \times p \times t$

22 $a \times m \times m$

5.04 Expressions from words

Easy Write an expression.

23 n minus 10

24 11 less than b

25 3 more than p

Medium Write an expression.

26 the product of a and x , plus 4

27 9 less than twice p

28 5 less than twice a

5.05 Substitution

Easy Substitute, then evaluate.

29 $p + 2$, when $p = 2$

30 $8p$, when $p = 12$

31 $p - 9$, when $p = 12$

Medium Substitute, then evaluate. Take care with negatives.

32 $5x + 1$, $x = 3$

33 $x + 8$, $x = 6$

34 $10 - x$, $x = 4$

35 $10 - x$, $x = -2$

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 36–58

5.07 Adding and subtracting terms

Easy Simplify by adding or subtracting like terms.

36 $2k - 10k$

37 $10y - 9y$

38 $2y - 8y$

39 $4x + 9x$

Medium Simplify. Collect the like terms.

40 $7x + 6y + 6x$

41 $7x + 3y + 4x + 4$

42 $8a - 2k - 7a$

5.08 Multiplying terms

Easy Simplify. Multiply the numbers, then the letters.

43 $5a \times 8c$

44 $3b \times 9p$

45 $5 \times 7k$

46 $2 \times 2n$

Medium Simplify.

47 $5py \times 6y$

48 $3cn \times 2cn$

49 $3ax \times 2ax$

50 $4t \times 4pt$

5.09 Dividing terms

Easy Simplify.

51 $49c \div 7$

52 $45a \div 9$

53 $72y \div 9$

54 $27x \div 9$

Medium Simplify.

55 $45a^2t \div 5a$

56 $28n^2t \div 4n$

57 $42x^2 \div 6x$

58 $21np \div 3p$

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 59–68

59

Easy

Pair friendly numbers.

$25 \times 17 \times 4$ Lesson 5.01

61

Easy

Write an expression.

10 less than 4 times x Lesson 5.04

63

Easy

Simplify.

$6t + 4 + 2t$ Lesson 5.07

65

Easy

Simplify.

$28xy \div 7y$ Lesson 5.09

67

Medium

Simplify $\frac{16ab^2}{4b}$. Lesson 5.09

60

Easy

Simplify.

$c \times 7 \times d$ Lesson 5.03

62

Easy

If $x = 3$ and $y = 4$, evaluate.

$2x + 3y$ Lesson 5.05

64

Easy

Simplify.

$6x \times 3y$ Lesson 5.08

66

Medium

Evaluate $5 \times 23 \times 2$. Lesson 5.01

68

Medium

Simplify $4a \times 6ab$. Lesson 5.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 69–77

69 **Medium** Simplify $4a + 7 - a + 2$. Lesson 5.07

70 **Medium** If $x = 3$ and $y = 7$, evaluate $2x + y$.
Lesson 5.05

71 **Medium** Write “4 less than y ”. Lesson 5.04

72 **Medium** Simplify $4 \times p \times 3 \times q$. Lesson 5.03

73 **Medium** Evaluate 19×6 . Lesson 5.02

74 **Challenging** Evaluate $20 \times 13 \times 5$. Lesson 5.01

75 **Challenging** Simplify $-2x \times 3x \times (-y)$. Lesson 5.08

76 **Challenging**
Substitute into each formula.
If $x = -3$, evaluate $x^2 + 2x$. Lesson 5.05

77 **Challenging** Write $5a^2b$ in expanded form.
Lesson 5.03

YOU DO

On your own: extra working space. Write the question number next to your working.